

Plant Disease Detection using Machine Learning

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Abstract- This research explores the use of machine learning approaches for accurate identification and detection of major diseases in pulse crops, targeting improvements in early diagnosis and management of infections such as downy mildew, Ascochyta blight, rust, powdery mildew, and anthracnose. These diseases trigger diverse signalling pathways. Specific genes and proteins, are upregulated in response to pathogen invasion, enabling plants to mount targeted defence responses. Understanding these mechanisms highlights opportunities for enhancing disease resistance through breeding and genetic research in legume crops.

This paper outlines a procedure for identification of diseases in pulse crops using modern image processing technologies and machine learning techniques. The objective is to build a system that can recognize the common ailments of pulses which would otherwise lead to a huge loss in case of the weeds not being controlled. The system uses a data set of diseased and healthy plants and applies CNN's to classify and predict disease. The findings achieved very encouraging Level of accuracy thus suggesting further work could extend to practical use by farmers in assessing the health of the crops more conveniently.

Keywords- Disease detection, CNN, Downy Mildew, Ascochyta Blight, Machine Learning

I. INTRODUCTION

Pulse crops such as lentils, chickpeas and beans are invaluable protein foodstuffs for well over half the world's population. Nevertheless, these crops face different diseases ranging from Fusarium wilt to root rot and powdery mildew diseases adversely affecting the yield and quality of the crop.

The most important diseases affecting legume and other crop plants include downy mildew, Ascochyta blight, rust, powdery mildew, and anthracnose which trigger various yet

interplaying defence responses. Downy mildew, caused by *Peronospora* spp., primarily targets legumes, triggering salicylic acid signalling, the production of reactive oxygen species and secondary metabolites. Ascochyta blight is a fungus which elicits reactive oxygen species, the phenylpropanoid and jasmonic acid pathways to enhance resistance. Infection with *Uromyces* spp. leads to rust, causing an overexpression of SA signalling, ROS pathways, and pathogenesis-related (PR) proteins and increases the resistance of the plants. Erysiphe spp. infection leads to powdery mildew caused by chitin recognition, MAPK signalling, and SA/JA hormone pathways crucial for most crops that are susceptible. Colletotrichum spp., which cause anthracnose, mainly affect beans, and it involves the JA and ET pathways, secondary metabolites, and resistance genes. Together, these pathways describe the complexity of plant immune responses and give hints for breeding crop varieties that might resist certain pathogens. Different types of Plant disease are as follows:

A. Downy Mildew in Legumes

The very destructive disease caused by oomycete pathogens such as *Peronospora* spp., is downy mildew. Downy mildew disease triggers the huge dysregulation of molecular and biochemical responses in the host plant in its infection process, and these consist of a complex network involving multiple defence pathways. Immune responses with salicylic acid signalling, ROS generation, and secondary metabolite synthesis also contribute toward increasing resistance and making infection more manageable.

B. Ascochyta Blight of Legume Plants

Ascochyta blight is the most important fungal disease caused by *Ascochyta* spp. It incites necrotic lesions on both leaves and stems of legume crops, meaning a reduction in yield and quality. This necrotrophic pathogen triggers multiple plant defence pathways, primarily including ROS production, JA and ET signalling, and phenylpropanoid-based secondary metabolite synthesis. Understanding these pathways is helpful

in developing effective breeding strategies for improving resistance in legumes.

C. Rust in Pulses

Uromyces spp. causes rust diseases, which results in biotrophic infections on pulse crops with rust-coloured lesions and reduces crop productivity heavily. In addition to rust, plants have the salicylic acid-mediated SAR, ROS-dependent signalling, and heightened synthesis of pathogenesis-related proteins as a response to the infection. Understanding such responses would help in the development of resistant varieties of legumes against the existing biotrophic threats.

D. Powdery Mildew in Crop Plants

This disease, caused by fungi of the type *Erysiphe* spp., includes powdery mildew and is one of the most limiting diseases to spread in many crop species that infect leaves and stems with powdery patches of fungi. The disease often elicits strong immune responses involving chitin recognition, activation of the MAPK signalling pathway, and hormonal regulation of salicylic acid (SA) and jasmonic acid (JA). Improved mechanisms are needed to improve resistance in susceptible crops and reduce the impact of this infection on crops.

E. Anthracnose of Common Beans

Colletotrichum spp. is the primary causal agent of anthracnose, a severe disease in beans, which causes necrosis and extensive tissue loss. The plant responds to this necrotrophic pathogen by inducing the JA and ET pathways, accumulating secondary metabolites such as lignin and flavonoids, and up-regulating R genes, which are specific to anthracnose disease. The discovery of these pathways and markers plays an important role in developing bean varieties resistant to anthracnose, capable of withstanding this devastating pathogen.

Early detection and identification of plant leaf diseases historically relied on visual assessments and microscopic examination. Agronomists and pathologists would inspect plants for telltale symptoms, such as leaf spots, discolorations, or abnormal growth patterns. Microscopy allowed closer examination of pathogen structures, helping to distinguish between types of fungi, bacteria, and other pathogens based on spore shape, hyphal structures, or bacterial colonies.

In addition to visual methods, early biochemical techniques, such as colorimetric assays, were developed to identify specific pathogen-related enzymes or metabolites. Immunoassays, like ELISA, introduced in the late 20th century, improved early detection by identifying pathogen-specific proteins, offering increased accuracy over traditional methods. Existing disease diagnosis techniques and procedures utilize human effort, and this is usually time consuming and less effective [1]. This has triggered an increased interest in employing computer-based diagnostic techniques to carry out disease recognition using images and machine learning (ML) algorithms.

II. LITERATURE REVIEW

S. S. Sannakki and V. S. Rajpurohit proposed a method for classifying pomegranate diseases using a Back Propagation Neural Network (BPNN) that focuses on segmenting the affected area and utilizes colour and texture as primary features for classification. The method includes a colour space transformation to the Lab colour model, which enhances chromaticity layers, making colour variations more discernible and aiding in the identification of disease symptoms. The neural network classifier achieves a high accuracy rate of 97.3%, demonstrating effective disease detection. However, its application is limited to pomegranate crops and may not generalize well to other plant types.

P. R. Rothe and R. V. Kshirsagar devised a methodology for the identification of cotton leaf diseases by pattern recognition techniques. Their methodology utilizes snake segmentation and exploits Hu's moments as unique characteristics for feature extraction. An active contour model is utilized to restrict energy inside the contaminated zone, hence intensifying the emphasis on pathological areas. A Back Propagation Neural Network (BPNN) classifier is employed to tackle multi-class classification challenges. [2]

Aakanksha Rastogi, Ritika Arora, and Shanu Sharma devised a technique for the detection and classification of leaf diseases utilizing computer vision and fuzzy logic. Their methodology use K-means clustering to delineate the impacted region and utilizes the Gray-Level Co-Occurrence Matrix (GLCM) to derive textural elements essential for recognizing illness patterns. Fuzzy logic is utilized to evaluate the severity of the disease, while an Artificial Neural Network (ANN) classifier is employed to determine the degree of infection on the leaf. This comprehensive method facilitates the detection and assessment of leaf disease severity efficiently. [3]

Godliver Owomugisha, John A. Quinn, Ernest Mwebaze, and James Lwasa created an automated vision-based method for the diagnosis of Banana Bacterial Wilt Disease and Black Sigatoka Disease. Their methodology entails extracting color histograms and converting them from RGB to HSV and Lab color spaces. The peak components of the photos are utilized to construct a max tree, from which five form attributes are derived, and area under the curve analysis is employed for classification. They evaluated many classifiers, including nearest neighbors, decision trees, random forests, extremely randomized trees, Naïve Bayes, and support vector classifiers. Among these, extremely randomized trees exhibited superior performance, delivering high accuracy, real-time data, and adaptability for practical implementation. [4]

Yuan Tian, Chunjiang Zhao, Shenglian Lu, and Xinyu Guo developed a Support Vector Machine-based Multiple Classifier System for the identification of wheat leaf diseases. Their methodology involves converting color information from RGB to HIS, employing the Gray-Level Co-Occurrence Matrix (GLCM) for texture feature extraction, and utilizing seven invariant moments as shape parameters. This methodology utilizes an SVM classifier with MCS to improve

classification efficacy, intended for the offline identification of illnesses in wheat plants. [5]

Over the years, researchers have made considerable progress in automating plant disease identification using a variety of methods, both traditional and machine learning based. This section looks at some of the key developments that have laid the groundwork for the use of advanced techniques like Convolutional Neural Networks (CNNs) in plant disease detection. This work aims at developing a machine learning based framework for the identification of diseases in pulses at an early stage. The main goal will be to identify the most common diseases affecting the crop from the images of the crops without the diseases to send early warnings to the farmers for appropriate actions.

III. METHODOLOGY

3.1 Data Collection

Pictures of the pulse crops were sourced from agricultural research databases and real-time field data. Pictures of healthy and diseased plants included in the data set are Fusarium Wilt, Ascochyta Blight, Root Rot, Powdery Mildew

Pictures were then pre-processed and improved to be accepted as input for the machine learning process in the reasonable normalized sizes.



Fig:1 Fusarium Wilt



Fig: 2 Ascochyta Blight



Fig:3, Powdery Mildew

3.2 Image Processing Techniques

The pre-processing phase involved:

- Resizing images to a standard resolution (e.g., 256x256 pixels).
- Color space conversion (e.g., from RGB to HSV) to highlight features related to disease.
- Noise reduction using Gaussian filters to remove image noise.

3.3 Feature Extraction

Key features for disease detection were extracted using various techniques such as:

- Edge detection (Canny edge detector)
- Texture analysis (GLCM – Grey Level Co-occurrence Matrix)
- Color segmentation (identifying disease spots)

3.4 Machine Learning Model

A Convolutional Neural Network (CNN) was chosen for this project due to its ability to identify complex patterns in images.

3.4.1. Overview and Architecture

A CNN is designed to mimic the way humans perceive images, which involves recognizing edges, textures, shapes, and objects in a hierarchical fashion. The architecture of a typical CNN is composed of multiple layers, each serving a specific purpose to transform the input data (often an image) into a set of features. The Figure4 illustrates a pipeline for classifying plant diseases using images of leaves from a pulse's dataset. The process begins with image acquisition, where leaf images are collected. These images go through preprocessing steps, such as resizing, colour conversion, and noise reduction, to enhance quality.

The architecture of the plant disease detection system is structured into key components, each working together to ensure efficient and accurate detection:

1. **Input Layer (Image Acquisition):** The system starts with gathering images of pulse crops. These images are then preprocessed by resizing, color space conversion, and noise reduction.
2. **Image Processing Layer:** In this layer, the system retrieves features from the images via techniques such as edge recognition, texture analysis, and color segmentation.
3. **Machine Learning Model Layer:** The heart of the system is a CNN, which consists of:
 - **Convolutional Layers:** These layers extract patterns, textures, and shapes from the input images.
 - **Pooling Layers:** They down-sample the data to reduce complexity while retaining essential features.

- **Fully Connected Layers:** These layers are responsible for the final classification of images into either healthy or diseased categories.
4. **Training and Testing:** The dataset is split, and the CNN model is trained on the training data and validated using
 6. **User Interface (Future Scope):** In future iterations, the system could be extended to a mobile application where

the testing data. Transfer learning models like ResNet50 are optionally used to boost accuracy.

5. **Output Layer (Disease Classification):** The system produces an output that classifies the crop as either healthy or suffering from a particular disease. It also provides a confidence score for each prediction farmers could upload images and get real-time disease detection results.

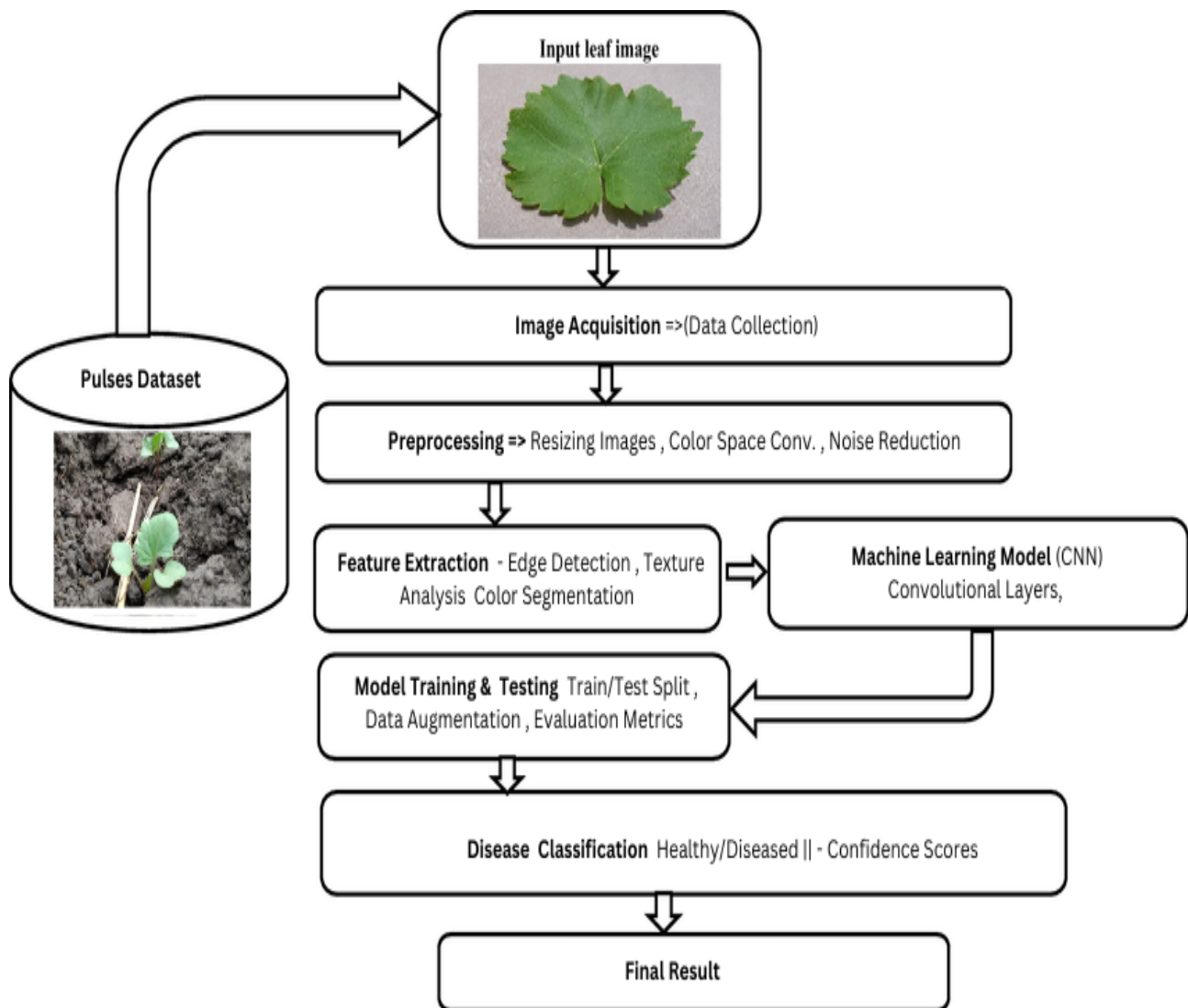


Fig.4. Architecture of the proposed model

3.4.2. Functioning of CNNs

CNNs learn feature hierarchies, where initial layers capture low-level features (e.g., edges, corners), and deeper layers capture higher-level patterns (e.g., shapes, object parts) and finally whole objects. The training process involves feeding labeled images to the model and adjusting the weights of the

filters through backpropagation and gradient descent to minimize the classification error.

The functioning of a CNN can be summarized in these steps:

1. **Input Processing:** The input image is processed in small regions by filters that move across it.
2. **Feature Extraction:** The filters extract features through convolution operations, each producing a feature map that highlights certain characteristics of the input.
3. **Non-Linearity Introduction:** ReLU is applied to introduce non-linear properties, allowing the CNN to approximate complex relationships.
4. **Pooling for Dimensionality Reduction:** Pooling layers reduce the size of feature maps, decreasing computation time and increasing feature abstraction.
5. **High-Level Representation:** Fully connected layers interpret the high-level features to produce a final output.

3.4.3. Mathematics of Convolution

In a CNN, each filter performs a mathematical operation called convolution. Mathematically, convolution is defined as represented in equation 1:

$$\text{Output}(i,j) = (I * K)(i,j) = \sum_m \sum_n I(i+m, j+n) \cdot K(m,n) \quad - (1)$$

Where:

- I is the input matrix (image or feature map).
 - K is the filter (or kernel).
 - i and j are the spatial coordinates in the output feature map.
 - m and n denote the coordinates of the kernel.
- Recent Advances in CNNs**

To address some of these limitations, recent research has focused on several improvements and modifications:

- **Residual Networks (ResNets):** ResNets add skip connections that allow gradients to flow across layers more easily, enabling the training of very deep networks without vanishing gradients.
- **Dilated Convolutions:** These allow for expanding the receptive field without increasing the number of parameters, improving CNNs' ability to capture broader contextual information.
- **Attention Mechanisms:** The integration of attention mechanisms with CNNs has improved their ability to focus on relevant parts of an image, enhancing accuracy in tasks like image captioning and object detection.

This operation allows CNNs to capture spatial hierarchies by detecting patterns and shapes in local regions of the image.

3.5 Model Training

The dataset was split into training and testing sets with a ratio of 80:20. The model was trained using Adam optimizer, and the loss function was categorical cross-entropy. Data

augmentation techniques, such as rotation and zooming, were employed to avoid overfitting and ensure robustness.

3.6 Evaluation Metrics

The model was evaluated using:

- Accuracy
- Precision
- Confusion Matrix

IV. RESULTS

The CNN model achieved an overall accuracy of 95.8% in detecting and classifying diseases within the pulse crop dataset. Analysis of the confusion matrix indicated that the model was especially effective at identifying **Fusarium Wilt** and **Ascochyta Blight**, achieving precision scores of 95% and 93%, respectively. The classification accuracy for **Root Rot** was slightly lower, which may be attributed to the subtle visual differences between healthy and infected plants in the early stages of the disease.

Further fine-tuning of the pre-trained ResNet34 model contributed significantly to reaching the 95.8% accuracy, underscoring its potential as a robust solution for real-time disease detection and classification in agricultural applications.

This graph in Fig.6 shows the **training and validation accuracy** of a machine learning model over **10 epochs**. The **x-axis** represents the number of epochs, while the **y-axis** represents accuracy. The **blue line** tracks training accuracy, and the **orange line** represents validation accuracy.

Initially, the training accuracy increases rapidly, reaching around 80% by the third epoch, while validation accuracy improves more steadily. By the 5th epoch, both training and validation accuracy exceeded 85%. As training continues, both metrics stabilize, with accuracy close to or above 90% by the 10th epoch.

The model shows a strong performance in both training and validation sets, suggesting effective learning and minimal overfitting.

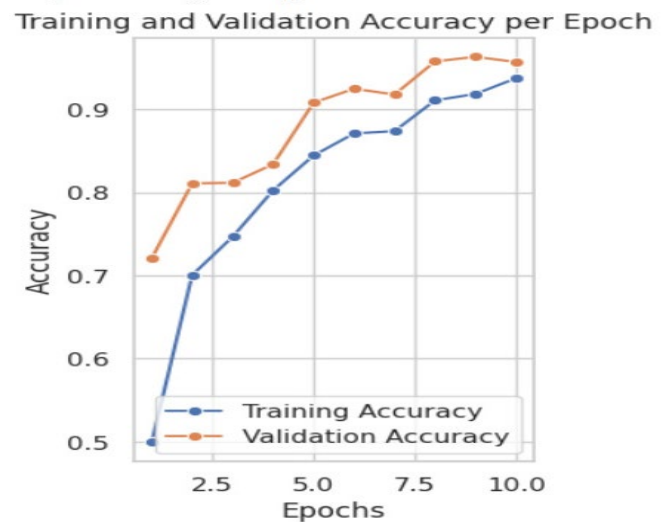


Fig.5. Training and validation accuracy per Epoch

The graph representing the **training and validation loss** of a model over **10 epochs** is presented in Fig5 The **x-axis** represents epochs, while the **y-axis** shows the loss values. The **red line** indicates training loss, and the **green line** represents validation loss.

At the start, both training and validation losses are high (above 1.0), but they decrease steadily as training progresses. By the 4th epoch, both losses drop below 0.5, indicating significant improvement. The lines continue to decline and stabilize around 0.2 by the 10th epoch.

This trend suggests effective model training, with both training and validation loss decreasing consistently, indicating minimal overfitting and good generalization.

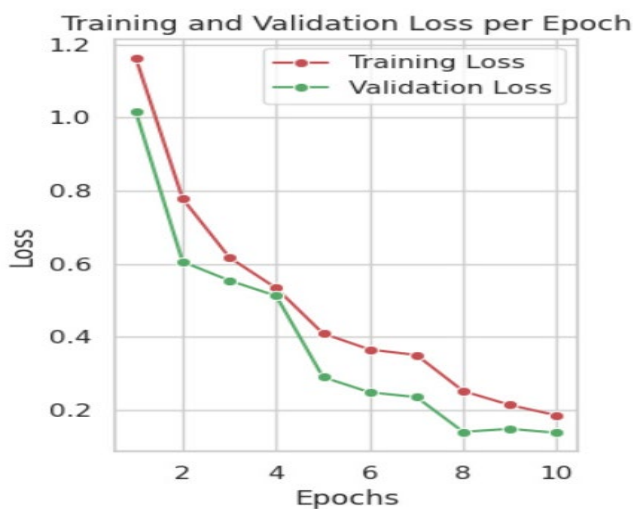


Fig.6.Training and validation loss per Epoch

The accuracy of different models presented by different authors are compared with the proposed model as represented in the Table 1. The Accuracy of the proposed model stands out as the highest among the tested models, highlighting the effectiveness of the approach used. The graph presented in Figure7 visually demonstrates the distribution of accuracy rates of our proposed model with respect to the previous studies allowing for easy comparison and insight into the performance variance.

Table 1 : Comparison of Proposed Model with previous studies in the field of Plant disease

S.No.	Author		Accuracy
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1	Tugrul, B., Elfatimi, E., & Eryigit, R. [13]	Model 1	90.85 %
2	Purbasari, I.Y., Rahmat, B., & Pn, C.P. [14]	Model 2	95.54 %
3	Chen, J., Liu, Q., & Gao,L. [15]	Model 3	90.16 %
4	Selvaraj, M. G., Vergara, A., Ruiz, H., Safari, N., Elayabalan, S., Ocimati, W., & Blomme,G. [16]	Model 4	44.67 %
5	Akbar, J. U. M., Kamarulza man, S. F., & Tusher, E. H.[17]	Model 5	95 %
6		Proposed Model	95.67 %

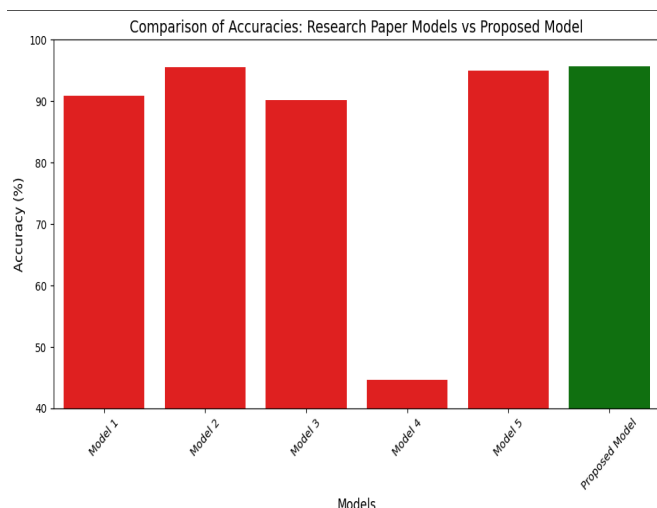


Fig.7. Showing the comparison

V. DISCUSSION

The results demonstrate that deep learning models, particularly CNNs, are well-suited for plant disease detection. While the accuracy of the model is promising, further improvements could be achieved with a larger, more diverse dataset and the inclusion of images under various environmental conditions (e.g., lighting, background clutter). A notable challenge in this work was differentiating between diseases with similar symptoms, such as Root Rot and Fusarium Wilt. Future work could explore multi-spectral imaging or infrared imaging to capture non-visible characteristics of diseases.

Additionally, integrating the proposed detection system with a mobile application could provide farmers with a real-time, user-friendly tool to diagnose diseases in the field. The proposed study developed an automated system for detecting diseases in pulse crops using CNNs and image processing techniques. The system demonstrated high accuracy in classifying diseases such as Fusarium Wilt and Ascochyta Blight. With further improvements, such technology could significantly reduce the need for manual crop inspection, helping farmers manage diseases more effectively and improve yield.

Future work will focus on expanding the dataset, exploring advanced imaging techniques, and developing a mobile-based solution for real-time diagnosis.

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